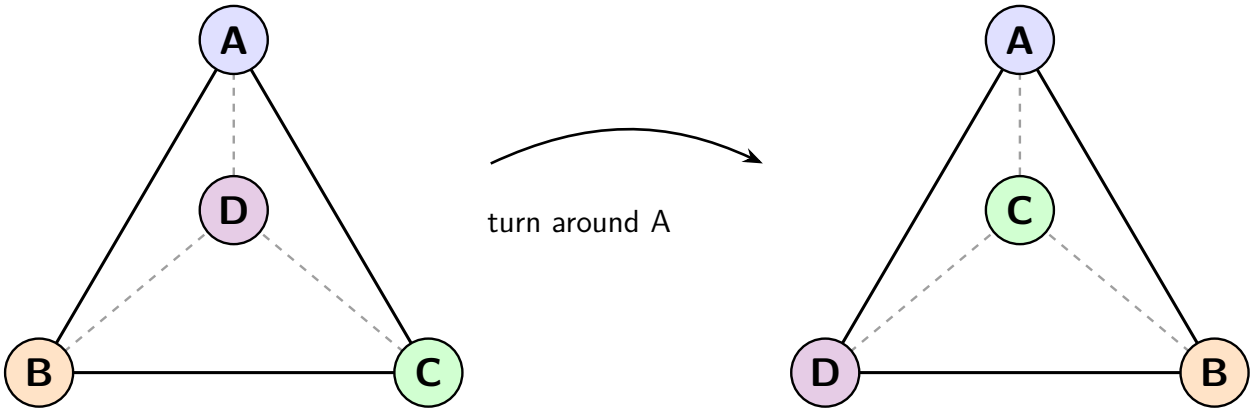
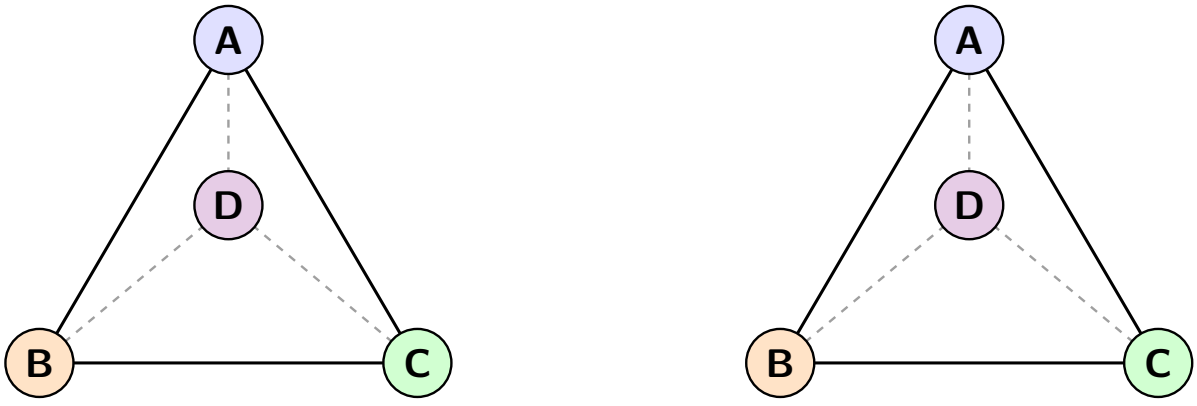
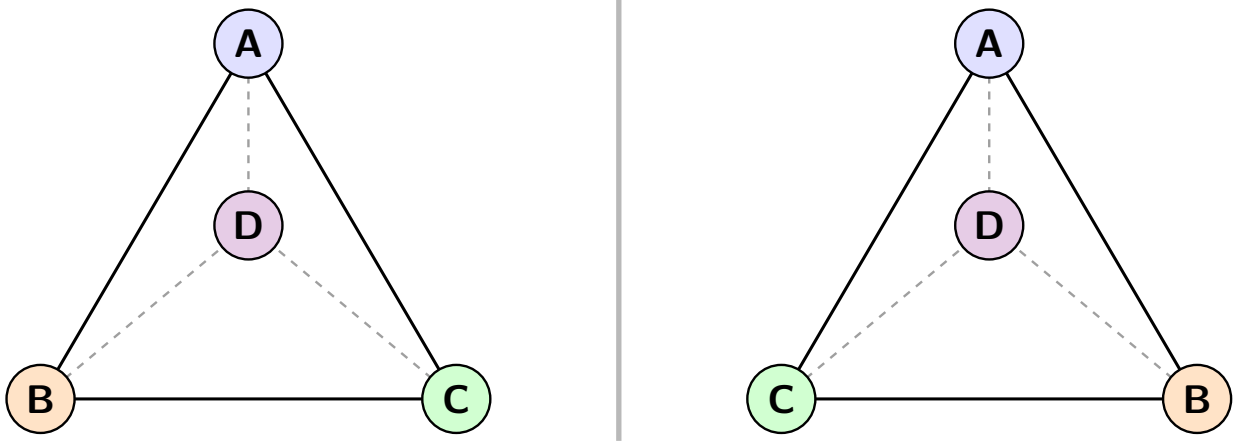
Use the rigid models. Move and turn them so same-letter corners match. Which way a letter points does not count. Keep the markers fixed unless a problem says to change them. Dashed edges pass behind the front face.



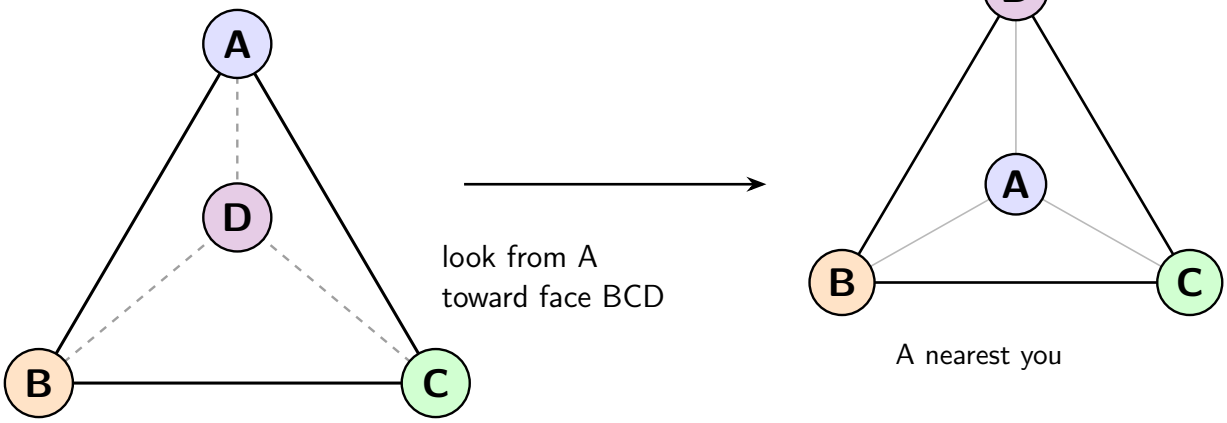
Problem 1: Build two copies of the first model. Turn one without changing any labels, and challenge a partner to match it.



Problem 2: Build the model and its mirror twin. Can one be moved and turned to match the other? If not, explain why no turn can.

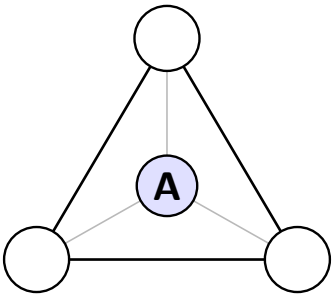


Problem 3: Look straight from A toward the center of the opposite face, with A nearest your eye. Use this view of both models from Problem 2 to explain whether any turn can match them.

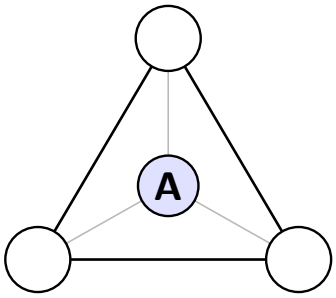


model

view from A

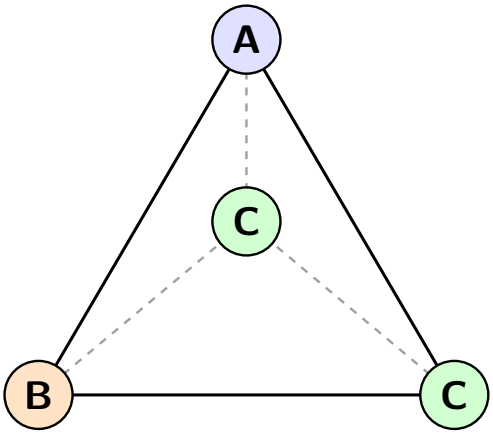


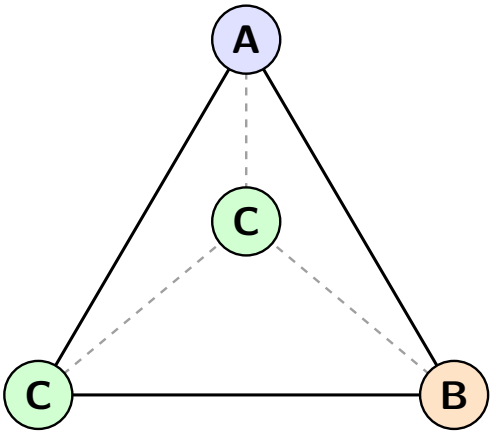
A nearest you



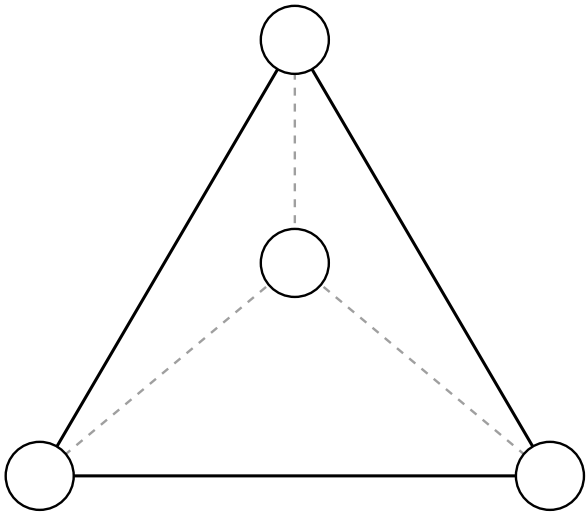
A nearest you

Problem 4: Choose two letters in the mirror pair and make them equal on both models. Start each trial with the original A, B, C, D mirror pair. Which choices allow a match by turning? Explain why your conclusion covers every choice.





Problem 5: How many rotations put a bare model back onto the same four positions? Count leaving it still as one. Turns count as the same when every starting corner ends at the same destination. Explain why your count is complete.



Problem 6: Put A, B, C and D on the model. How many different models can be made if models that match by turning count as the same? Explain.